

INF1004 Mathematics II - Solutions

Tutorial 4: Eigenvalues & Eigenvectors

SIT

1. Solution for Matrix A

$$A = \begin{pmatrix} 4 & 2 \\ 3 & 3 \end{pmatrix}$$

Find Eigenvalues: $\det(A - \lambda I) = 0$

$$\det \begin{pmatrix} 4 - \lambda & 2 \\ 3 & 3 - \lambda \end{pmatrix} = (4 - \lambda)(3 - \lambda) - 6 = 0$$

$$\lambda^2 - 7\lambda + 12 - 6 = 0 \implies \lambda^2 - 7\lambda + 6 = 0$$

$$(\lambda - 1)(\lambda - 6) = 0 \implies \boxed{\lambda_1 = 1, \lambda_2 = 6}$$

Find Eigenvectors: $(A - \lambda I)\vec{v} = 0$

• For $\lambda_1 = 1$: $\begin{pmatrix} 3 & 2 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \implies 3v_1 + 2v_2 = 0 \implies v_2 = -\frac{3}{2}v_1$ Let $v_1 = 2$,

then $v_2 = -3$. $\vec{v}_1 = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$

• For $\lambda_2 = 6$: $\begin{pmatrix} -2 & 2 \\ 3 & -3 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \implies -2v_1 + 2v_2 = 0 \implies v_1 = v_2$ Let $v_1 = 1$,

then $v_2 = 1$. $\vec{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

1. Solution for Matrix B

$$B = \begin{pmatrix} 7 & -3 \\ -2 & 2 \end{pmatrix}$$

Eigenvalues:

$$\det \begin{pmatrix} 7 - \lambda & -3 \\ -2 & 2 - \lambda \end{pmatrix} = (7 - \lambda)(2 - \lambda) - 6 = 0$$

$$\lambda^2 - 9\lambda + 14 - 6 = 0 \implies \lambda^2 - 9\lambda + 8 = 0$$

$$(\lambda - 1)(\lambda - 8) = 0 \implies \boxed{\lambda_1 = 1, \lambda_2 = 8}$$

Eigenvectors:

- $\lambda_1 = 1 : \begin{pmatrix} 6 & -3 \\ -2 & 1 \end{pmatrix} \vec{v} = 0 \implies 6v_1 - 3v_2 = 0 \implies v_2 = 2v_1.$

$$\text{Let } v_1 = 1 \implies \vec{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

- $\lambda_2 = 8 : \begin{pmatrix} -1 & -3 \\ -2 & -6 \end{pmatrix} \vec{v} = 0 \implies -v_1 - 3v_2 = 0 \implies v_1 = -3v_2.$

$$\text{Let } v_2 = 1 \implies \vec{v}_2 = \begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

1. Solution for Matrix C

$$C = \begin{pmatrix} 3 & 1 \\ -2 & 0 \end{pmatrix}$$

Eigenvalues:

$$\det \begin{pmatrix} 3-\lambda & 1 \\ -2 & -\lambda \end{pmatrix} = (3-\lambda)(-\lambda) + 2 = 0$$

$$\lambda^2 - 3\lambda + 2 = 0 \implies (\lambda - 1)(\lambda - 2) = 0 \implies \boxed{\lambda_1 = 1, \lambda_2 = 2}$$

Eigenvectors:

- $\lambda_1 = 1 : \begin{pmatrix} 2 & 1 \\ -2 & -1 \end{pmatrix} \vec{v} = 0 \implies 2v_1 + v_2 = 0 \implies v_2 = -2v_1.$

$$\vec{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

- $\lambda_2 = 2 : \begin{pmatrix} 1 & 1 \\ -2 & -2 \end{pmatrix} \vec{v} = 0 \implies v_1 + v_2 = 0 \implies v_2 = -v_1.$

$$\vec{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

1. Solution for Matrix D

$$D = \begin{pmatrix} -2 & -2 \\ -5 & 1 \end{pmatrix}$$

Eigenvalues:

$$\det \begin{pmatrix} -2 - \lambda & -2 \\ -5 & 1 - \lambda \end{pmatrix} = (-2 - \lambda)(1 - \lambda) - 10 = 0$$

$$\lambda^2 + \lambda - 2 - 10 = 0 \implies \lambda^2 + \lambda - 12 = 0$$

$$(\lambda - 3)(\lambda + 4) = 0 \implies \boxed{\lambda_1 = 3, \lambda_2 = -4}$$

Eigenvectors:

- $\lambda_1 = 3 : \begin{pmatrix} -5 & -2 \\ -5 & -2 \end{pmatrix} \vec{v} = 0 \implies -5v_1 - 2v_2 = 0 \implies v_2 = -\frac{5}{2}v_1.$

$$\text{Let } v_1 = 2 \implies \vec{v}_1 = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$$

- $\lambda_2 = -4 : \begin{pmatrix} 2 & -2 \\ -5 & 5 \end{pmatrix} \vec{v} = 0 \implies 2v_1 - 2v_2 = 0 \implies v_2 = v_1.$

$$\vec{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

2. Diagonalize Matrix A

$$A = \begin{pmatrix} 4 & 5 \\ -1 & -2 \end{pmatrix}$$

1. Eigenvalues: $(4 - \lambda)(-2 - \lambda) + 5 = \lambda^2 - 2\lambda - 3 = (\lambda + 1)(\lambda - 3) = 0$.

$$\lambda_1 = -1, \quad \lambda_2 = 3$$

2. Eigenvectors:

$$\bullet \lambda_1 = -1: \begin{pmatrix} 5 & 5 \\ -1 & -1 \end{pmatrix} \vec{v} = 0 \implies v_1 + v_2 = 0 \implies \vec{v}_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\bullet \lambda_2 = 3: \begin{pmatrix} 1 & 5 \\ -1 & -5 \end{pmatrix} \vec{v} = 0 \implies v_1 + 5v_2 = 0 \implies \vec{v}_2 = \begin{pmatrix} -5 \\ 1 \end{pmatrix}$$

3. Matrices P and D:

$$P = \begin{pmatrix} 1 & -5 \\ -1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}$$

4. Find P^{-1} : $\det(P) = (1)(1) - (-5)(-1) = -4$. $P_{adj} = \begin{pmatrix} 1 & 5 \\ 1 & 1 \end{pmatrix}$.

Diagonalize Matrix A contd..

$$A = PDP^{-1} = \begin{pmatrix} 1 & -5 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix} \frac{1}{-4} \begin{pmatrix} 1 & 5 \\ 1 & 1 \end{pmatrix}$$

2. Diagonalize Matrix B

$$B = \begin{pmatrix} 7 & -2 \\ -1 & 8 \end{pmatrix}$$

Eigenvalues: $(7 - \lambda)(8 - \lambda) - 2 = \lambda^2 - 15\lambda + 54 = (\lambda - 6)(\lambda - 9) = 0$.

$$\lambda_1 = 6, \quad \lambda_2 = 9$$

Eigenvectors:

- $\lambda_1 = 6 : \begin{pmatrix} 1 & -2 \\ -1 & 2 \end{pmatrix} \vec{v} = 0 \implies v_1 = 2v_2 \implies \vec{v}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

- $\lambda_2 = 9 : \begin{pmatrix} -2 & -2 \\ -1 & -1 \end{pmatrix} \vec{v} = 0 \implies v_1 = -v_2 \implies \vec{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

Diagonalization:

$$P = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 6 & 0 \\ 0 & 9 \end{pmatrix}$$

$$\det(P) = -2 - 1 = -3.$$

$$B = PDP^{-1} = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 6 & 0 \\ 0 & 9 \end{pmatrix} \frac{1}{-3} \begin{pmatrix} -1 & -1 \\ -1 & 2 \end{pmatrix}$$

3. Matrix A - Eigenvalues

$$A = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 2 & 0 \\ 3 & 1 & -3 \end{pmatrix}$$

Characteristic Equation: $\det(A - \lambda I) = 0$

$$\det \begin{pmatrix} 1 - \lambda & 0 & 4 \\ 0 & 2 - \lambda & 0 \\ 3 & 1 & -3 - \lambda \end{pmatrix} = 0$$

Expand along row 2:

$$(2 - \lambda) [(1 - \lambda)(-3 - \lambda) - (4)(3)] = 0$$

$$(2 - \lambda) [\lambda^2 + 2\lambda - 3 - 12] = 0$$

$$(2 - \lambda)(\lambda^2 + 2\lambda - 15) = 0$$

$$(2 - \lambda)(\lambda + 5)(\lambda - 3) = 0$$

$$\boxed{\lambda_1 = 2, \quad \lambda_2 = 3, \quad \lambda_3 = -5}$$

3. Matrix A - Eigenvectors

1. For $\lambda_1 = 2$: $\begin{pmatrix} -1 & 0 & 4 \\ 0 & 0 & 0 \\ 3 & 1 & -5 \end{pmatrix} \vec{v} = 0$

$$-v_1 + 4v_3 = 0 \implies v_1 = 4v_3$$

$$3v_1 + v_2 - 5v_3 = 0 \implies 3(4v_3) + v_2 - 5v_3 = 0 \implies v_2 = -7v_3$$

Let $v_3 = 1 \implies \vec{v}_1 = \begin{pmatrix} 4 \\ -7 \\ 1 \end{pmatrix}$

2. For $\lambda_2 = 3$: $\begin{pmatrix} -2 & 0 & 4 \\ 0 & -1 & 0 \\ 3 & 1 & -6 \end{pmatrix} \vec{v} = 0$

$$-2v_1 + 4v_3 = 0 \implies v_1 = 2v_3$$

$$-v_2 = 0 \implies v_2 = 0$$

Let $v_3 = 1 \implies \vec{v}_2 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$

3. Matrix A - Eigenvectors P Matrix

3. For $\lambda_3 = -5$: $\begin{pmatrix} 6 & 0 & 4 \\ 0 & 7 & 0 \\ 3 & 1 & 2 \end{pmatrix} \vec{v} = 0$

$$6v_1 + 4v_3 = 0 \implies v_3 = -\frac{3}{2}v_1$$

$$7v_2 = 0 \implies v_2 = 0$$

Let $v_1 = 2 \implies v_3 = -3$. $\vec{v}_3 = \begin{pmatrix} 2 \\ 0 \\ -3 \end{pmatrix}$

Construct P and D:

$$P = \begin{pmatrix} 4 & 2 & 2 \\ -7 & 0 & 0 \\ 1 & 1 & -3 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -5 \end{pmatrix}$$

3. Matrix A - Inverse P (Adjoint Method)

Calculate $P^{-1} = \frac{1}{\det(P)} P_{adj}$

$$\det(P) = 4(0) - 2(21) + 2(-7) = -42 - 14 = -56.$$

Cofactor Matrix (C):

$$C = \begin{pmatrix} + \begin{vmatrix} 0 & 0 \\ 1 & -3 \end{vmatrix} & - \begin{vmatrix} -7 & 0 \\ 1 & -3 \end{vmatrix} & + \begin{vmatrix} -7 & 0 \\ 1 & 1 \end{vmatrix} \\ - \begin{vmatrix} 2 & 2 \\ 1 & -3 \end{vmatrix} & + \begin{vmatrix} 4 & 2 \\ 1 & -3 \end{vmatrix} & - \begin{vmatrix} 4 & 2 \\ 1 & 1 \end{vmatrix} \\ + \begin{vmatrix} 2 & 2 \\ 0 & 0 \end{vmatrix} & - \begin{vmatrix} 4 & 2 \\ -7 & 0 \end{vmatrix} & + \begin{vmatrix} 4 & 2 \\ -7 & 0 \end{vmatrix} \end{pmatrix} = \begin{pmatrix} 0 & -21 & -7 \\ 8 & -14 & -2 \\ 0 & -14 & 14 \end{pmatrix}$$

$$P_{adj} = C^T = \begin{pmatrix} 0 & 8 & 0 \\ -21 & -14 & -14 \\ -7 & -2 & 14 \end{pmatrix}$$

$$P^{-1} = \frac{1}{-56} \begin{pmatrix} 0 & 8 & 0 \\ -21 & -14 & -14 \\ -7 & -2 & 14 \end{pmatrix}$$

Answer: $A = PDP^{-1}$ (Verified)

4. Verification and Eigenvalues

$$A = \begin{pmatrix} -1 & 1 & 2 \\ -6 & 2 & 6 \\ 0 & 1 & 1 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Verify $\vec{v}$:

$$A\vec{v} = \begin{pmatrix} -1 + 1 + 2 \\ -6 + 2 + 6 \\ 0 + 1 + 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

So, $\vec{v}$ is an eigenvector with eigenvalue $\lambda_1 = 2$.

Find other eigenvalues: $\det(A - \lambda I) = 0$

$$(-1 - \lambda)(2 - \lambda)(1 - \lambda) + 1(6)(0) + 2(-6)(1) - [0 + 1(6)(-1 - \lambda) + (1 - \lambda)(-6)] = 0$$

Simplifying leads to:

$$(-1 - \lambda)(2 - \lambda)(1 - \lambda) = 0$$

$$\boxed{\lambda_1 = 2, \quad \lambda_2 = 1, \quad \lambda_3 = -1}$$

4. Finding Other Eigenvectors

$$\text{For } \lambda_2 = 1: \begin{pmatrix} -2 & 1 & 2 \\ -6 & 1 & 6 \\ 0 & 1 & 0 \end{pmatrix} \vec{v} = 0$$

$$0v_1 + 1v_2 + 0v_3 = 0 \implies v_2 = 0$$

$$-2v_1 + 2v_3 = 0 \implies v_1 = v_3$$

$$\text{Let } v_1 = 1 \implies \vec{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$\text{For } \lambda_3 = -1: \begin{pmatrix} 0 & 1 & 2 \\ -6 & 3 & 6 \\ 0 & 1 & 2 \end{pmatrix} \vec{v} = 0$$

$$v_2 + 2v_3 = 0 \implies v_2 = -2v_3$$

$$-6v_1 + 3v_2 + 6v_3 = 0 \implies -6v_1 + 3(-2v_3) + 6v_3 = 0 \implies -6v_1 = 0 \implies v_1 = 0$$

$$\text{Let } v_3 = 1 \implies \vec{v}_3 = \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}$$

4. Diagonalization

$$P = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & -2 \\ 1 & 1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Find P^{-1} : $\det(P) = 1(2) - 1(3) + 0 = -1$.

Calculating Cofactors and Transpose (P_{adj}):

$$P_{adj} = \begin{pmatrix} 2 & -1 & -2 \\ -3 & 1 & 2 \\ 1 & 0 & -1 \end{pmatrix}$$

$$P^{-1} = \frac{1}{-1} \begin{pmatrix} 2 & -1 & -2 \\ -3 & 1 & 2 \\ 1 & 0 & -1 \end{pmatrix} = \begin{pmatrix} -2 & 1 & 2 \\ 3 & -1 & -2 \\ -1 & 0 & 1 \end{pmatrix}$$

$$A = PDP^{-1}$$

5. Verification and Eigenvalues

$$A = \begin{pmatrix} -5 & 8 & 32 \\ 2 & 1 & -8 \\ -2 & 2 & 11 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$$

1. Verify $\vec{v}$ is an eigenvector:

$$A\vec{v} = \begin{pmatrix} -5(2) + 8(-2) + 32(1) \\ 2(2) + 1(-2) - 8(1) \\ -2(2) + 2(-2) + 11(1) \end{pmatrix} = \begin{pmatrix} 6 \\ -6 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$$

Since $A\vec{v} = 3\vec{v}$, $\vec{v}$ is an eigenvector with eigenvalue $\lambda_1 = 3$.

2. Characteristic Equation:

$$\det(A - \lambda I) = (-5 - \lambda)(1 - \lambda)(11 - \lambda) + \dots = 0$$

Simplifies to: $(1 - \lambda)(\lambda - 3)(\lambda - 3) = 0$

$$\boxed{\lambda_{1,2} = 3, \quad \lambda_3 = 1}$$

5. Eigenvectors for $\lambda = 3$

Solve $(A - 3I)\vec{x} = 0$:

$$\begin{pmatrix} -8 & 8 & 32 \\ 2 & -2 & -8 \\ -2 & 2 & 8 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

From row 2: $2x_1 - 2x_2 - 8x_3 = 0 \implies x_1 - x_2 - 4x_3 = 0 \implies x_1 = x_2 + 4x_3$.

General solution with free variables x_2, x_3 :

$$\vec{x} = \begin{pmatrix} x_2 + 4x_3 \\ x_2 \\ x_3 \end{pmatrix} = x_2 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$$

Selection for P: We use the verified vector $\vec{v}_1$ and one basic vector $\vec{v}_2$:

$$\vec{v}_1 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} \text{ (Given), } \vec{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \text{ (Derived from } x_2 = 1, x_3 = 0 \text{)}$$

(Note: If we let $x_2 = 0, x_3 = 1$, $\vec{v}_2 = (4, 0, 1)^T$ is an alternative). Diagonalization not unique, go to slide 20 for this alternative solution.

5. Eigenvector for $\lambda = 1$

Solve $(A - 1I)\vec{x} = 0$:

$$\begin{pmatrix} -6 & 8 & 32 \\ 2 & 0 & -8 \\ -2 & 2 & 10 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

From row 2: $2x_1 - 8x_3 = 0 \implies x_1 = 4x_3$. From row 3: $-2x_1 + 2x_2 + 10x_3 = 0$.
Substitute x_1 :

$$-2(4x_3) + 2x_2 + 10x_3 = 0 \implies -8x_3 + 2x_2 + 10x_3 = 0$$

$$2x_2 = -2x_3 \implies x_2 = -x_3$$

Let $x_3 = 1$:

$$\vec{v}_3 = \begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}$$

5. Diagonalization

Construct P and D :

$$P = \begin{pmatrix} 2 & 1 & 4 \\ -2 & 1 & -1 \\ 1 & 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Inverse of P : $\det(P) = 2(1) - 1(-1) + 4(-1) = 2 + 1 - 4 = -1$.

$$P_{adj} = \begin{pmatrix} 1 & -1 & -5 \\ 1 & -2 & -6 \\ -1 & 1 & 4 \end{pmatrix}$$

$$P^{-1} = \frac{1}{-1} P_{adj} = \begin{pmatrix} -1 & 1 & 5 \\ -1 & 2 & 6 \\ 1 & -1 & -4 \end{pmatrix}$$

Verify: $PDP^{-1} = A$.

5. Alternative Selection for $\lambda = 3$

Alternative Choice: From the general solution for $\lambda = 3$:

$$\vec{x} = x_2 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$$

Instead of mixing the given vector with the first basis vector, we can simply choose the second basis vector (setting $x_2 = 0, x_3 = 1$): $\vec{v}_2 = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$

New Diagonalization Matrices: Using $\vec{v}_1$ (Given), $\vec{v}_2$ (Alternative), and $\vec{v}_3$ ($\lambda = 1$):

$$P = \begin{pmatrix} 2 & 4 & 4 \\ -2 & 0 & -1 \\ 1 & 1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Inverse of Alternative P:

$$P = \begin{pmatrix} 2 & 4 & 4 \\ -2 & 0 & -1 \\ 1 & 1 & 1 \end{pmatrix}, \quad P^{-1} = \begin{pmatrix} -0.5 & 0 & 2 \\ -0.5 & 1 & 3 \\ 1 & -1 & -4 \end{pmatrix}$$

Solution 6: The "Mini-Internet"

1. Set up the Equation ($M\vec{r} = \vec{r}$):

$$(M - I)\vec{r} = \vec{0} \implies \begin{pmatrix} 0.5 - 1 & 1 \\ 0.5 & 0 - 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$
$$\begin{pmatrix} -0.5 & 1 \\ 0.5 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

2. Solve the System:

$$0.5x - y = 0 \implies y = 0.5x \implies x = 2y$$

Let $y = t$, then $x = 2t$

$$\vec{r}_{\text{raw}} = \begin{pmatrix} 2t \\ t \end{pmatrix}$$

3. Normalize (Total = 100%): Total = $2t + t = 3t$.

$$\vec{r}_{\text{final}} = \begin{pmatrix} 2t/3t \\ t/3t \end{pmatrix} = \begin{pmatrix} 2/3 \\ 1/3 \end{pmatrix} \approx \begin{pmatrix} 66.7\% \\ 33.3\% \end{pmatrix}$$

Conclusion: Site A is twice as popular as Site B. Probability of visiting it is $2/3$ and that of visiting site B is $1/3$.